Case Study 2 - Gasoline Vs. Alternative Fuel Vehicles

MECH 408 - Applied Engineering Economy
Colorado State University

Group 2
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Introduction

Consumers generally prioritize financial over nonfinancial when making personal transportation decisions. This study evaluates five vehicle types—conventional gasoline, alternative fuel (E85), hybrid (HEV), plug-in hybrid (PHEV), and battery electric (BEV)—based on purely financial factors including upfront and repeating costs, as well as end-of-life value and any government incentives.

A month-based economic model is used to make the variety of costs and benefits associated with these different vehicle types comparable, and through Monte Carlo simulations the uncertainty of these costs is quantified to further inform the monetary worth of each vehicle type.

Methodology

For this analysis it was decided to identify a four-door sedan type vehicle of model year 2024 for each vehicle family. For the calculations it was also assumed that each vehicle would have both a six-year lifespan and identical miles driven per week. The cost factors considered in the analysis are periodic fixed costs including maintenance, fuel, and registration fees, the costs incurred from loans subtracting any government incentives, and one-time costs such as home charger installation and salvage or resale value.

The monthly loan payment was calculated assuming an 11%monthly compounding interest rate over a 6 year period, with the original loan amount being the full MSRP of the vehicle in question minus any applicable state and/or federal incentives. This was calculated using *Equation 1*, where *i* is the interest rate per month and *n* is the number of months over which the loan is paid.

$$Monthly\ Loan\ Amount = (MSRP - Tax\ Credit)\frac{i*(1+i)^n}{(1+i)^n-1} \quad (1)$$

To calculate the annual expenditure on fuel, the annual distance travel was calculated first. It was assumed that a uniform distribution of 50 miles/day 240-260 days/year would be representative of mandatory travel. Travel for recreation was taken as a normal distribution of about 100 miles/trip with a standard deviation of 10 miles, with a lognormal distribution of trips/year with a minimum of 16 and a maximum of 24. The result is calculated with *Equation 2* below.

$$Miles\ Per\ Year = \left(\frac{miles}{day} * \frac{days}{year}\right)_{mandatory} + \left(\frac{miles}{trip} * \frac{trips}{year}\right)_{recreation} \quad (2)$$

Research for the estimated range of each vehicle was then performed with an uncertainty accounted for by adding a lognormal distribution to each value of -60 minimum and +20 maximum. Next the cost/unit fuel nominal value, lower, and upper bounds were estimated from different sources for each type of fuel in Colorado, with the resultant values shown in *Table 1* below.

Table 1: Estimated Cost in $/gal or $/kWh

Fuel Type	Nominal	Distribution	
		Minimum	Maximum
Gasoline	3.30	3.10	4.00
E85	2.50	2.20	3.10
Electricity	0.15	0.14	0.18

Finally, for each vehicle the capacity of fuel held in each 'tank' was found, and all values were substituted in Equation *3* seen below.

$$Fuel\ Cost\ per\ Year = \left(\frac{T}{R}\right) * (\Sigma_{1,2\ldots}(C_{1,2\ldots} * c_{1,2\ldots})) \quad (3)$$

Where *T* is the annual miles traveled(miles/year), *R* is the range(miles/tank), *C* is the fuel capacity(unit fuel/tank), *c* is the fuel cost($/unit fuel), and the subscript indicates the fuel type.

From research the estimated cost of the installation of a home charger for PHEV and BEV electrical vehicles was found as $1,200 and $3,000 respectively, with this cost being considered an upfront cost and applied at month 0 of the model.

The annual cost of maintenance was spread over two appointments per year, with research yielding the estimates seen in *Table 2* below.

Table 2: Annual Maintenance Cost and Distribution($/Year)

Vehicle Type	Maintenance Cost	Lognormal Distribution	
		Min	Max
Gasoline	500	450	800
E85	550	500	850
HEV	600	550	850
OHEV	600	550	850
BEV	400	350	800

The salvage value was found using an estimate of a baseline End of Life value of 35% of the MSRP at 6 years, then applied a normal distribution with a standard deviation of 10% to the calculated End of Life value to account for variances due to low/high offers and minor damage.

The annual vehicle registration fee was calculated from Colorado DoT's resources wherein various road fees were compiled with the specific ownership tax(SOT) added on top. The SOT is a percentage of the taxable value of a vehicle(85% of MSRP), with the percentage being 2.1% the first year, 1.5% the second, 1.2% the third, 0.9% the fourth, and .45% the fifth and sixth years.

With the values from the above processes, the present worth is calculated for each month that the vehicle is owned using *Equation 4* below.Where n is the specific month the present worth is being calculated for, A is the sum of costs for that month, and i is the annual discount rate(estimated 6%) divided by 12. The results for all 72 months of the simulation are then summed to arrive at the Net Present Value which is used to compare the options.

$$Present\ Worth = \frac{A}{(1+i)^n} \qquad (4)$$

Results and Analysis

Table 3 below lists the costs of ownership for the different vehicle models by their type.

Table 3: Comparisons

Category	Gasoline (Toyota Corolla LE)	E85 Flex-Fuel (Chevy Malibu LT)	HEV (Honda Accord Hybrid)	PHEV (Hyundai Sonata Plug-in)	BEV (Tesla Model 3 RWD)
MSRP / Price	$22,000	$28,000	$32,000	$34,000	$39,000
Average Monthly Loan Payment	~$415	~$528	~$603	~$641	~$735
Annual Fuel Cost	~$1,400	~$1,000	~$1,000	~$1,500	~$400
Annual Maintenance	~$500	~$550	~$600	~$600	~$400
Battery	N/A	N/A	$2k–$8k (often	$3k–$10k	$12k–$16k

Replacement			covered)	(warranty)	(warranty)
Charger Install Cost	N/A	N/A	N/A	~$1,200	~$3,000
6-Yr Salvage	~$7,700	~$9,800	~$11,200	~$11,900	~$13,650
Incentives	None	None	Not eligible	Not eligible	Fed: up to $7,500 + $3,500 State

In *Figure 1* below, it can be observed that the median vehicle cost is $31.9k, or functionally identical to the HEV vehicle(Honda Accord Hybrid)'s MSRP.

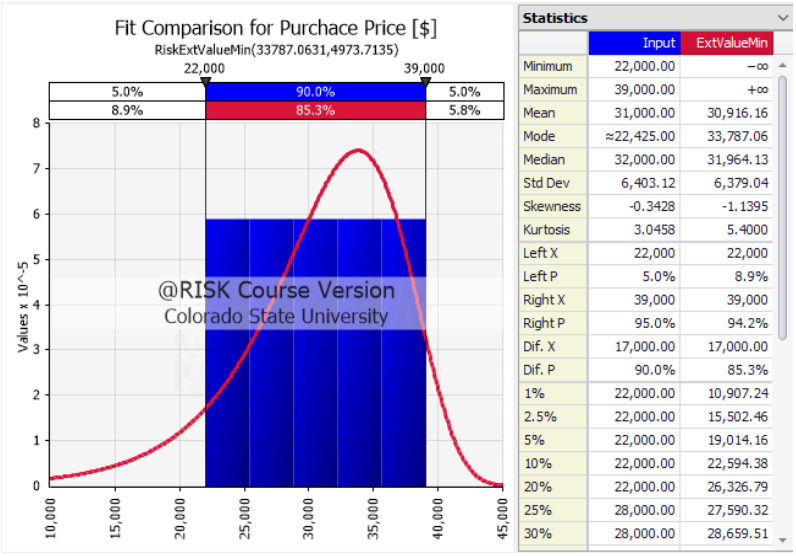


Figure 1: @RISK - MSRP Distribution

Table 4 below shows the percentage probability that different options would have the lowest NPC from our Monte-Carlo simulation based on the probability distributions assigned to each input as described in the methodologies section previously. For the full results of the simulation, see *Appendix B:@Risk -NPC Distribution*

Table 4: Net Present Cost Outcome Summary

Vehicle Type	Median NPC	Standard Deviation	Probability Lowest Cost
Gas(Toyota Corolla)	$31,686	±$1036	90.00%
E85(Chevy Malibu)	$35,531	±$1064	1.40%
HEV(Honda Accord Hybrid)	$40,624	±$1160	0.00%
PHEV(Hyundai Sonata)	$45,345	±$1332	0%
BEV(Tesla Mod 3)	$32,469	±$1155	8.60%

Conclusion and Recommendation

The Monte Carlo Simulations performed on the input variables show that the traditional gasoline fuel vehicle(Toyota Corolla) is financially superior in a supermajority of scenarios. This can be seen in *Table 4* above wherein this option has a 90% probability of having the lowest Net Present Cost, and it can be seen in greater detail in *Appendix B:@RISK - NPC Distribution*. However, as can be seen in *Appendix A:@RISK - Fuel Price Distribution*, the price of fuel is an input with a wide variance, with electricity acting as a 'financial anchor' that keeps the NPC low for the BEV and hybrid vehicles. Thus if it is expected that fuel prices will remain very high, then the battery electric vehicle will become the ideal choice in terms of the net present cost.

However, this does not mean that the traditional gasoline vehicle is not the best option. Only in 8.6% of scenarios is the BEV choice superior, and for the vast majority of people the societal or environmental benefits that are felt indirectly are outweighed by the benefits felt directly in the form of the long term savings from the gasoline-fueled vehicle.

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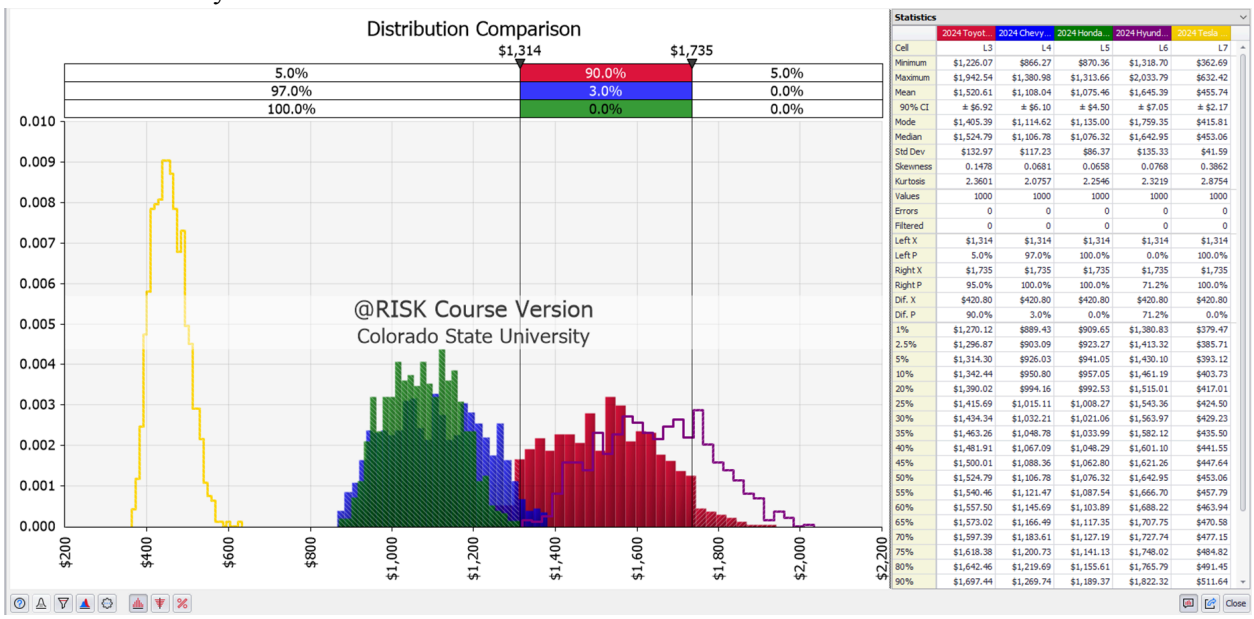
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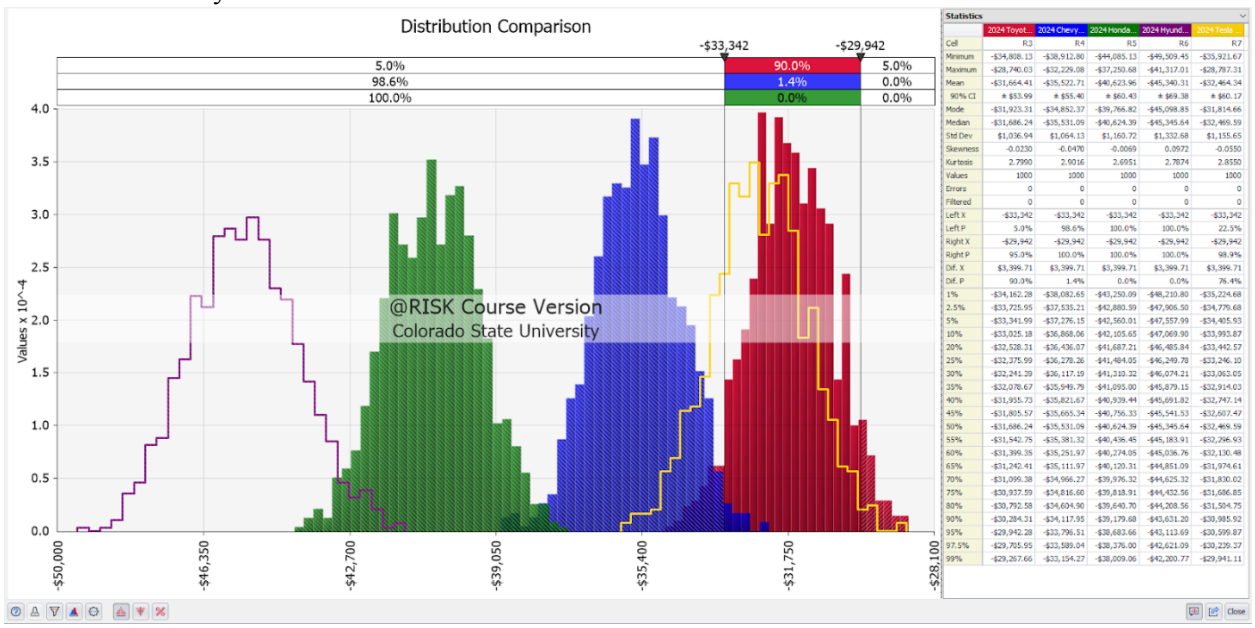
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Appendix A: @RISK - Fuel Price Distribution
Monte Carlo Analysis:



	2024 Toyot...	2024 Chevy...	2024 Honda...	2024 Hyund...	2024 Tesla
Cell	L3	L4	L5	L6	L7
Minimum	$1,226.07	$866.27	$870.36	$1,318.70	$362.69
Maximum	$1,942.54	$1,380.98	$1,313.66	$2,033.79	$632.42
Mean	$1,520.61	$1,108.04	$1,075.46	$1,645.39	$455.74
90% CI	± $6.92	± $6.10	± $4.50	± $7.05	± $2.17
Mode	$1,405.39	$1,114.62	$1,135.00	$1,759.35	$415.81
Median	$1,524.79	$1,106.78	$1,076.32	$1,642.95	$453.06
Std Dev	$132.97	$117.23	$86.37	$135.33	$41.59
Skewness	0.1478	0.0681	0.0658	0.0768	0.3862
Kurtosis	2.3601	2.0757	2.2546	2.3219	2.8754
Values	1000	1000	1000	1000	1000
Errors	0	0	0	0	0
Filtered	0	0	0	0	0
Left X	$1,314	$1,314	$1,314	$1,314	$1,314
Left P	5.0%	97.0%	100.0%	0.0%	100.0%
Right X	$1,735	$1,735	$1,735	$1,735	$1,735
Right P	95.0%	100.0%	100.0%	71.2%	100.0%
Dif. X	$420.80	$420.80	$420.80	$420.80	$420.80
Dif. P	90.0%	3.0%	0.0%	71.2%	0.0%
1%	$1,270.12	$889.43	$909.65	$1,380.83	$379.47
2.5%	$1,296.87	$903.09	$923.27	$1,413.32	$385.71
5%	$1,314.30	$926.03	$941.05	$1,430.10	$393.12
10%	$1,342.44	$950.80	$957.05	$1,461.19	$403.73
20%	$1,390.02	$994.16	$992.53	$1,515.01	$417.01
25%	$1,415.69	$1,015.11	$1,008.27	$1,543.36	$424.50
30%	$1,434.34	$1,032.21	$1,021.06	$1,563.97	$429.23
35%	$1,463.26	$1,048.78	$1,033.99	$1,582.12	$435.50
40%	$1,481.91	$1,067.09	$1,048.29	$1,601.10	$441.55
45%	$1,500.01	$1,088.36	$1,062.80	$1,621.26	$447.64
50%	$1,524.79	$1,106.78	$1,076.32	$1,642.95	$453.06
55%	$1,540.46	$1,121.47	$1,087.54	$1,666.70	$457.79
60%	$1,557.50	$1,145.69	$1,103.89	$1,688.22	$463.94
65%	$1,573.02	$1,166.49	$1,117.35	$1,707.75	$470.58
70%	$1,597.39	$1,183.61	$1,127.19	$1,727.74	$477.15
75%	$1,618.38	$1,200.73	$1,141.13	$1,748.02	$484.82
80%	$1,642.46	$1,219.69	$1,155.61	$1,765.79	$491.45
90%	$1,697.44	$1,269.74	$1,189.37	$1,822.32	$511.64

Appendix B: @RISK - NPC Distribution
Monte Carlo Analysis:



	2024 Toyot...	2024 Chevy...	2024 Honda...	2024 Hyund...	2024 Tesla
Cell	R3	R4	R5	R6	R7
Minimum	-$34,808.13	-$38,912.80	-$44,085.13	-$49,569.45	-$35,921.67
Maximum	-$28,740.03	-$32,229.08	-$37,250.68	-$41,317.01	-$28,787.31
Mean	-$31,664.41	-$35,522.71	-$40,623.96	-$45,340.31	-$32,464.34
90% CI	± $53.99	± $55.40	± $60.43	± $69.38	± $60.17
Mode	-$31,923.31	-$34,852.37	-$39,766.82	-$45,098.85	-$31,814.66
Median	-$31,686.24	-$35,531.09	-$40,624.39	-$45,345.64	-$32,469.59
Std Dev	$1,036.94	$1,064.13	$1,160.72	$1,332.68	$1,155.65
Skewness	-0.0230	-0.0470	-0.0069	0.0572	-0.0550
Kurtosis	2.7990	2.9016	2.6951	2.7874	2.8550
Values	1000	1000	1000	1000	1000
Errors	0	0	0	0	0
Filtered	0	0	0	0	0
Left X	-$33,342	-$33,342	-$33,342	-$33,342	-$33,342
Left P	5.0%	98.6%	100.0%	100.0%	22.5%
Right X	-$29,942	-$29,942	-$29,942	-$29,942	-$29,942
Right P	95.0%	100.0%	100.0%	100.0%	98.9%
Dif. X	$3,399.71	$3,399.71	$3,399.71	$3,399.71	$3,399.71
Dif. P	90.0%	1.4%	0.0%	0.0%	76.4%
1%	-$34,162.28	-$38,082.65	-$43,250.09	-$48,210.80	-$35,224.68
2.5%	-$33,725.95	-$37,535.21	-$42,880.59	-$47,906.50	-$34,779.68
5%	-$33,341.99	-$37,276.15	-$42,560.01	-$47,557.99	-$34,405.93
10%	-$33,025.18	-$36,868.06	-$42,105.65	-$47,069.90	-$33,993.87
20%	-$32,528.31	-$36,436.07	-$41,687.21	-$46,485.84	-$33,442.57
25%	-$32,375.99	-$36,278.26	-$41,484.05	-$46,249.78	-$33,246.10
30%	-$32,241.29	-$36,117.19	-$41,310.32	-$46,074.21	-$33,063.05
35%	-$32,078.67	-$35,949.79	-$41,095.00	-$45,879.15	-$32,914.03
40%	-$31,935.73	-$35,821.67	-$40,939.44	-$45,691.82	-$32,747.14
45%	-$31,805.57	-$35,665.34	-$40,756.33	-$45,541.53	-$32,607.47
50%	-$31,686.24	-$35,531.09	-$40,624.39	-$45,345.64	-$32,469.59
55%	-$31,542.75	-$35,381.32	-$40,439.45	-$45,183.91	-$32,296.93
60%	-$31,396.35	-$35,251.97	-$40,274.05	-$45,036.76	-$32,130.48
65%	-$31,242.41	-$35,111.97	-$40,120.31	-$44,851.09	-$31,974.61
70%	-$31,099.38	-$34,966.27	-$39,976.32	-$44,625.32	-$31,830.02
75%	-$30,937.59	-$34,816.60	-$39,818.91	-$44,432.56	-$31,686.85
80%	-$30,792.58	-$34,604.90	-$39,640.70	-$44,208.56	-$31,504.75
90%	-$30,284.31	-$34,117.95	-$39,179.68	-$43,631.20	-$30,985.92
95%	-$29,942.28	-$33,796.51	-$38,683.66	-$43,113.69	-$30,599.87
97.5%	-$29,705.95	-$33,589.04	-$38,376.00	-$42,621.09	-$30,226.27
99%	-$29,267.66	-$33,154.27	-$38,009.06	-$42,200.77	-$29,941.11